

Small-Scale InstructGPT: Reproducing RLHF with DistilGPT2

GOKULANATH MAHESH KUMAR, Cornell University, USA

AIDAN WU, Cornell University, USA

1 Introduction

Large language models trained only with next-token prediction often continue text fluently without reliably following user intent. Ouyang et al.’s *Training Language Models to Follow Instructions with Human Feedback* introduced InstructGPT, a three-stage alignment pipeline: supervised fine-tuning (SFT), reward modeling from human preference comparisons, and reinforcement learning with PPO. Its central contribution is that post-training alignment can make outputs more helpful and preferred by users, even when the aligned model is smaller than an unaligned base model.

We re-implemented this idea at course scale using DistilGPT2 instead of billion-parameter GPT models. Our goal was not to match OpenAI’s human-evaluation numbers exactly, but to test whether the same alignment pipeline changes behavior in the expected direction under limited compute and public datasets. We also added a Direct Preference Optimization (DPO) branch to compare PPO-based RLHF against reward-free preference optimization.

2 Chosen Result

We focused on the InstructGPT result that alignment can improve preference quality more than raw scale alone. In the paper’s headline human-evaluation comparison, reported in the model-size preference results (Figure 1 in Ouyang et al.), labelers prefer outputs from a 1.3B InstructGPT model over outputs from the 175B GPT-3 model despite the aligned model having over 100 times fewer parameters. This result matters because it argues that the training objective and feedback signal can be as important as model size for instruction following.

Our reproduction cannot recreate the original labeler study, but it can test the same causal story at smaller scale: starting from the same DistilGPT2 base model, does adding instruction demonstrations and preference optimization improve outputs over SFT alone? We translated this into a controlled comparison among SFT, PPO-aligned, and DPO-aligned DistilGPT2. Instead of human preference labels, we used a learned reward model and qualitative inspection to ask whether preference optimization increases reward scores, win rates, and visible instruction following.

3 Methodology

We implemented the pipeline in Colab using Hugging Face Transformers, TRL, PyTorch, and public Hugging Face datasets.

- **SFT.** We started from DistilGPT2 (82M parameters) and fine-tuned it on Stanford Alpaca instruction-response examples for 500 training steps.
- **Reward model.** We initialized a DistilGPT2 sequence classifier from the SFT checkpoint and trained it on 5,000 Anthropic Helpful-Harmless chosen/rejected pairs with a binary ranking objective.
- **PPO alignment.** The SFT model served as policy and reference model. Generated responses were scored by the reward model, then optimized with PPO using a KL penalty to limit drift from the SFT policy.
- **DPO extension.** We trained a manual DPO branch on UltraFeedback Binarized pairs, optimizing directly against chosen/rejected examples.
- **Evaluation.** We evaluated eight held-out prompts with reward-model scores, pairwise win rates, response lengths, and qualitative inspection.

Several implementation choices were driven by compute limits. DistilGPT2 was small enough to fine-tune repeatedly on a Colab GPU, but its base capabilities were far below the GPT-3 models used in the original paper. We therefore treated the experiment as a pipeline reproduction rather than a capability reproduction. SFT prompts used explicit instruction and response delimiters. The reward model reused the SFT checkpoint as a sequence classifier, and PPO used the SFT model as both policy initialization and reference model, with KL regularization intended to limit drift.

The DPO branch was added because the project proposal emphasized reproducing InstructGPT, but the notebook results showed that PPO was difficult to stabilize at small scale. PPO requires a separate reward model, online generation, reward scoring, and careful KL control. DPO uses preference pairs directly, so it served as an independent exploration

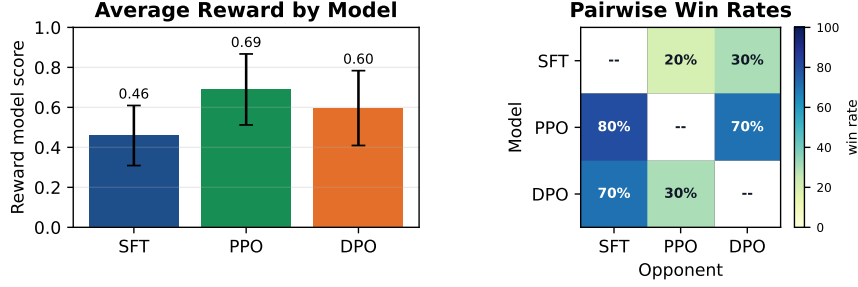


Fig. 1. Reward-model evaluation favored preference-tuned models. PPO achieved the highest average reward and beat SFT on 80% of prompts.

beyond the original paper’s PPO pipeline and a stress test of whether direct preference optimization was more stable in our small-data setting.

4 Results and Analysis

The intermediate training objectives improved, but alignment remained noisy. SFT reached a final loss of 2.3769, and the reward model reached a final loss of 0.6006. PPO completed 31 batches, but TRL repeatedly warned that KL divergence became negative, suggesting unstable reward optimization or generation settings. DPO was also noisy: sampled losses were 2.2272, 9.6528, 2.2586, 0.6931, and 4.0160, with a final reported loss of 3.6301.

Average reward scores were SFT 0.4592 ± 0.1502 , PPO 0.6896 ± 0.1776 , and DPO 0.5965 ± 0.1871 . PPO beat SFT on 80% of prompt-level reward comparisons and beat DPO on 70%; DPO beat SFT on 70%. Response length also increased after preference tuning: SFT averaged 35.8 words, PPO 47.9, and DPO 48.7. These results support the paper’s claim that feedback changes model behavior, while showing that reproducing the pipeline is not the same as reproducing the original scale of alignment quality. The reward model preferred PPO outputs, yet human inspection still found drift, hallucination, and repetition.

Qualitatively, the aligned models often looked more instruction-shaped but still failed semantically. On “staying healthy,” PPO produced more advice-like text, while DPO became repetitive. On factual prompts, all models hallucinated or gave incomplete explanations. On cake instructions, PPO and DPO were closer to the expected recipe format, but they still included irrelevant or underspecified details. This suggests that our alignment signal improved surface helpfulness before it improved factual reliability.

Prompt	SFT baseline	PPO/RLHF	DPO
Solar eclipse	Fluent but scientifically wrong; repeats sun imagery.	Longer and more confident, but still hallucinates visual details.	Very short and incorrect; attributes eclipse to a meteorite.
Staying healthy	Produces a list, but repeats “healthy” and exercise.	More natural advice about diet and hydration.	Long continuation with repeated phrases and little new content.
Bake a cake	Confuses cake with sauce/meat and gives bad timing.	More recipe-shaped, but includes irrelevant ingredients.	Shorter and closer to the task, but underspecified.

Table 1. Representative qualitative failures. Alignment improved style and structure more reliably than factuality.

The most important discrepancy from the original paper is evaluation scale. Ouyang et al. use large-scale human preferences and evaluate against strong GPT-3 baselines. Our reward model was trained on a small 5k-pair subset and reused as the primary evaluator, so high reward scores indicate agreement with our proxy model rather than definitive human preference. This explains why PPO can dominate the reward metric while still producing brittle outputs.

This mismatch is informative. PPO appeared best under the learned reward, yet its responses sometimes became verbose or overconfident. DPO avoided reward-model queries during training, but our short manual run had unstable losses and did not consistently outperform PPO. Overall, the results reproduce the direction of the InstructGPT finding

while showing that small-model RLHF can over-optimize shallow markers of helpfulness such as length, confidence, and format.

5 Reflections

RLHF behaved as a multi-stage system rather than a single training trick. SFT established instruction-response formatting, the reward model supplied a measurable preference signal, and PPO/DPO shifted outputs toward that signal. However, each stage introduced failure modes. DistilGPT2 has limited factual knowledge and coherence, so preference optimization often amplified style without reliably improving truthfulness. PPO was sensitive to KL behavior, while DPO was simpler conceptually but unstable in our short manual run.

This changed how we interpret “reproduction” for alignment papers. We could reproduce the mechanics of SFT, reward modeling, PPO, and DPO, but the behavior depended heavily on model capacity, preference data quality, and reward calibration. The project was therefore most useful as an implementation study: it exposed where RLHF obtains its signal and where small-scale versions become fragile.

The biggest lesson was that evaluation design matters as much as training design. Using the trained reward model as the evaluator created circularity: PPO was judged by the same kind of proxy signal that trained it. A stronger evaluation would separate training reward from validation preference, use held-out human or LLM-judge comparisons, and include task-specific checks for factuality and repetition.

Given more time, we would use a stronger instruction model with LoRA or QLoRA, evaluate on more prompts, add human or LLM-judge ratings, tune PPO KL settings, compare against a standard DPO implementation, and separate reward-model validation from the reward model used for PPO training.

References

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